

HANDS-ON 1 [Beginner] Schema Design & Core SQL — DDL and Normalisation

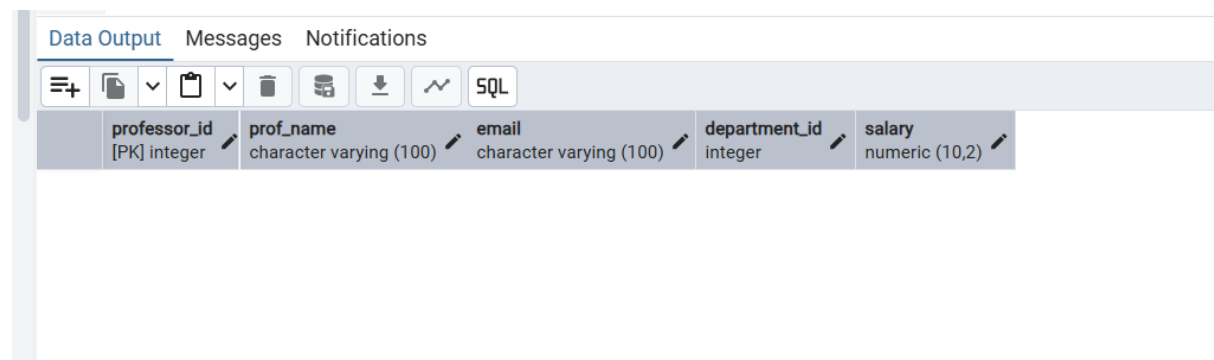
Task 1: Create the Database and Tables

1. Table Created



The screenshot shows a database management interface with three tabs: "Data Output", "Messages", and "Notifications". The "Messages" tab is selected. Below the tabs, the text "CREATE TABLE" is displayed. Further down, a status message reads "Query returned successfully in 955 msec."

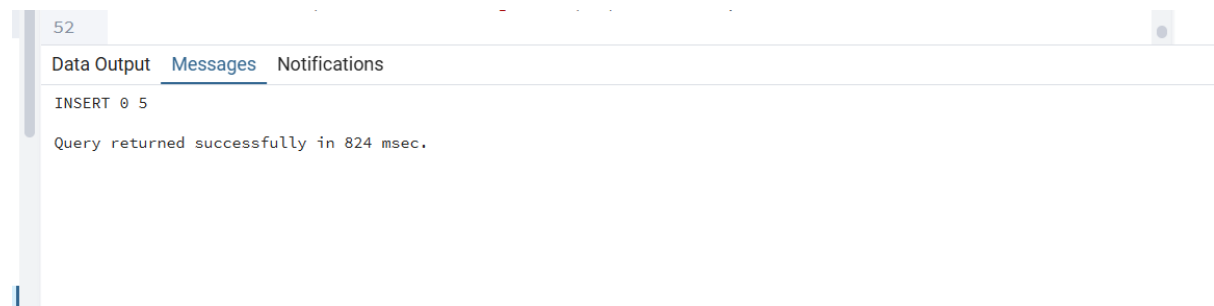
2. Retrieve empty tables as no data has been inserted yet



The screenshot shows a database management interface with three tabs: "Data Output", "Messages", and "Notifications". The "Data Output" tab is selected. Below the tabs is a toolbar with icons for various actions, including a table icon. Below the toolbar, the schema of a table is displayed with the following columns:

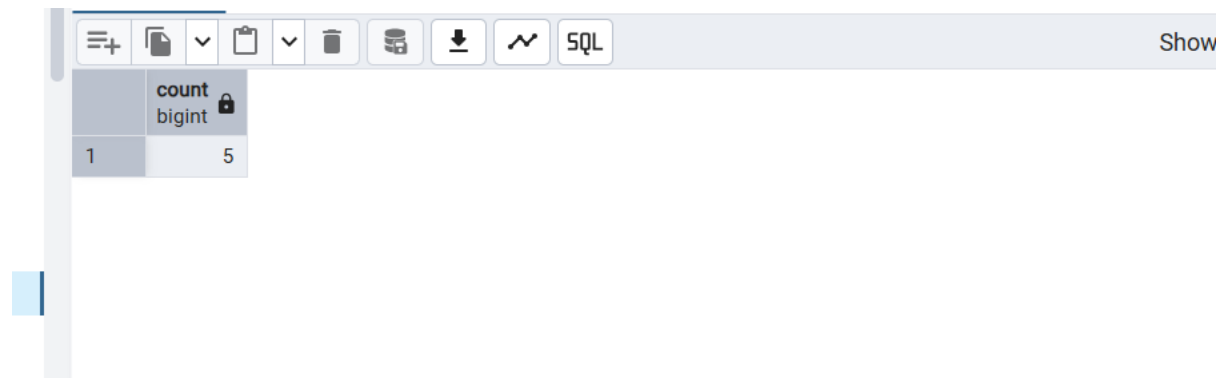
professor_id	prof_name	email	department_id	salary
[PK] integer	character varying (100)	character varying (100)	integer	numeric (10,2)

3. Insert sample data into the tables



The screenshot shows a database management interface with three tabs: "Data Output", "Messages", and "Notifications". The "Messages" tab is selected. Below the tabs, the text "INSERT 0 5" is displayed. Further down, a status message reads "Query returned successfully in 824 msec."

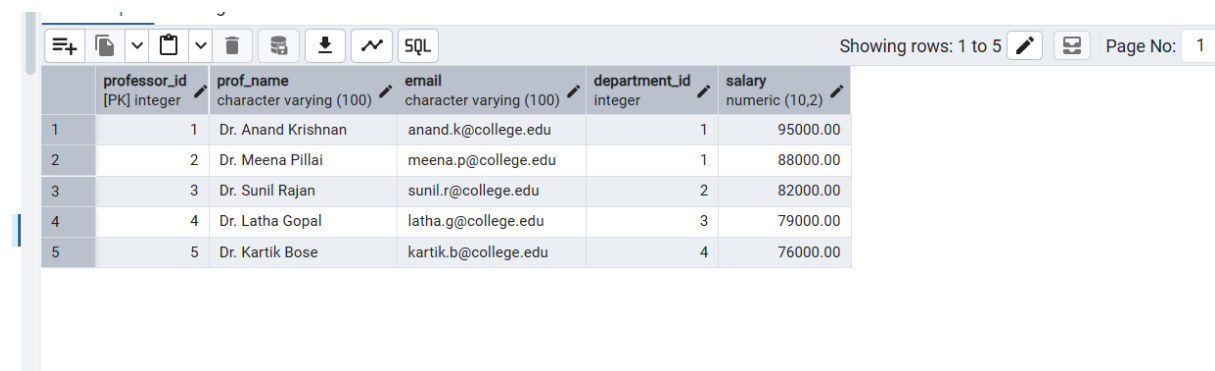
4. Check if all tables have been populated with data—Count of Courses has been retrieved



The screenshot shows a database management interface. At the top, there is a toolbar with icons for various database operations and a text input field containing 'SQL'. Below the toolbar, a table is displayed with the following structure:

	count
1	5

5.Retrieve all data from the tables to verify the inserted records—Table Professor has been retrieved



The screenshot shows a database management interface. At the top, there is a toolbar with icons for various database operations and a text input field containing 'SQL'. Below the toolbar, a table is displayed with the following structure:

	professor_id [PK] integer	prof_name character varying (100)	email character varying (100)	department_id integer	salary numeric (10,2)
1	1	Dr. Anand Krishnan	anand.k@college.edu	1	95000.00
2	2	Dr. Meena Pillai	meena.p@college.edu	1	88000.00
3	3	Dr. Sunil Rajan	sunil.r@college.edu	2	82000.00
4	4	Dr. Latha Gopal	latha.g@college.edu	3	79000.00
5	5	Dr. Kartik Bose	kartik.b@college.edu	4	76000.00

Task 2: Verify Normalisation

```
-- TASK 02:Normalization Analysis
-- 1NF (First Normal Form)
-- All tables contain atomic (single) values in every column.
-- There are no repeating groups or multiple values stored in one field.
-- For example, storing multiple phone numbers in one column like
'9876543210,9123456789'
-- would violate 1NF. Instead, each phone number should be stored separately.

-- 2NF (Second Normal Form)
-- Every non-key attribute is fully dependent on the primary key.
-- In the enrollments table, the logical candidate key is (student_id,
course_id).
-- The attribute grade depends on the complete student-course enrollment,
-- not on student_id or course_id individually. Hence the table satisfies 2NF.

-- 3NF (Third Normal Form)
-- There are no transitive dependencies in the schema.
```

```
-- Student details do not store department names directly.  
-- Instead, students reference department_id, and department details are  
-- stored  
-- only in the departments table.  
-- Therefore, the schema satisfies Third Normal Form (3NF).
```

Task 3: Alter and Extend the Schema

1. Add Phone Number to Students Table

Data Output

Messages

Notifications

ALTER TABLE

Query returned successfully in 275 msec.

2. verify the addition of new column

Data Output

Messages

Notifications

SQL

	column_name name
1	student_id
2	first_name
3	last_name
4	email
5	date_of_birth
6	department_id
7	enrollment_year
8	phone_number

3. Add max_seats to Courses Table

Data Output

Messages

Notifications

ALTER TABLE

Query returned successfully in 225 msec.

4. Verify the addition of the new column

vs

3

	column_name name
1	course_id
2	course_name
3	course_code
4	credits
5	department_id
6	max_seats

5. Add CHECK Constraints on grade

Data Output Messages Notifications

ALTER TABLE

Query returned successfully in 132 msec.

6. Verify the CHECK constraint

```
10 VALUES(1, 3, CURRENT_DATE, 'Z');
```

Data Output Messages Notifications

```
ERROR:  new row for relation "enrollments" violates check constraint "chk_grade"
Failing row contains (13, 1, 3, 2026-06-28, Z ).

SQL state: 23514
Detail: Failing row contains (13, 1, 3, 2026-06-28, Z ).
```

7. Rename hod_name to head_of_dept

```
7
Data Output Messages Notifications
ALTER TABLE
Query returned successfully in 110 msec.
```

8. Verify the renaming of the column

	column_name
	name
1	department_id
2	dept_name
3	head_of_dept
4	budget

9. Rollback the addition of the phone_number column in students table

```
13 WHERE table_name = 'students';
Data Output Messages Notifications
ALTER TABLE
Query returned successfully in 130 msec.
```

10. Verify the removal of the column

```
12 SELECT column_name FROM information_schema.c
13 WHERE table_name = 'students';
```

Data Output Messages Notifications



	column_name name
1	student_id
2	first_name
3	last_name
4	email
5	date_of_birth
6	department_id
7	enrollment_year